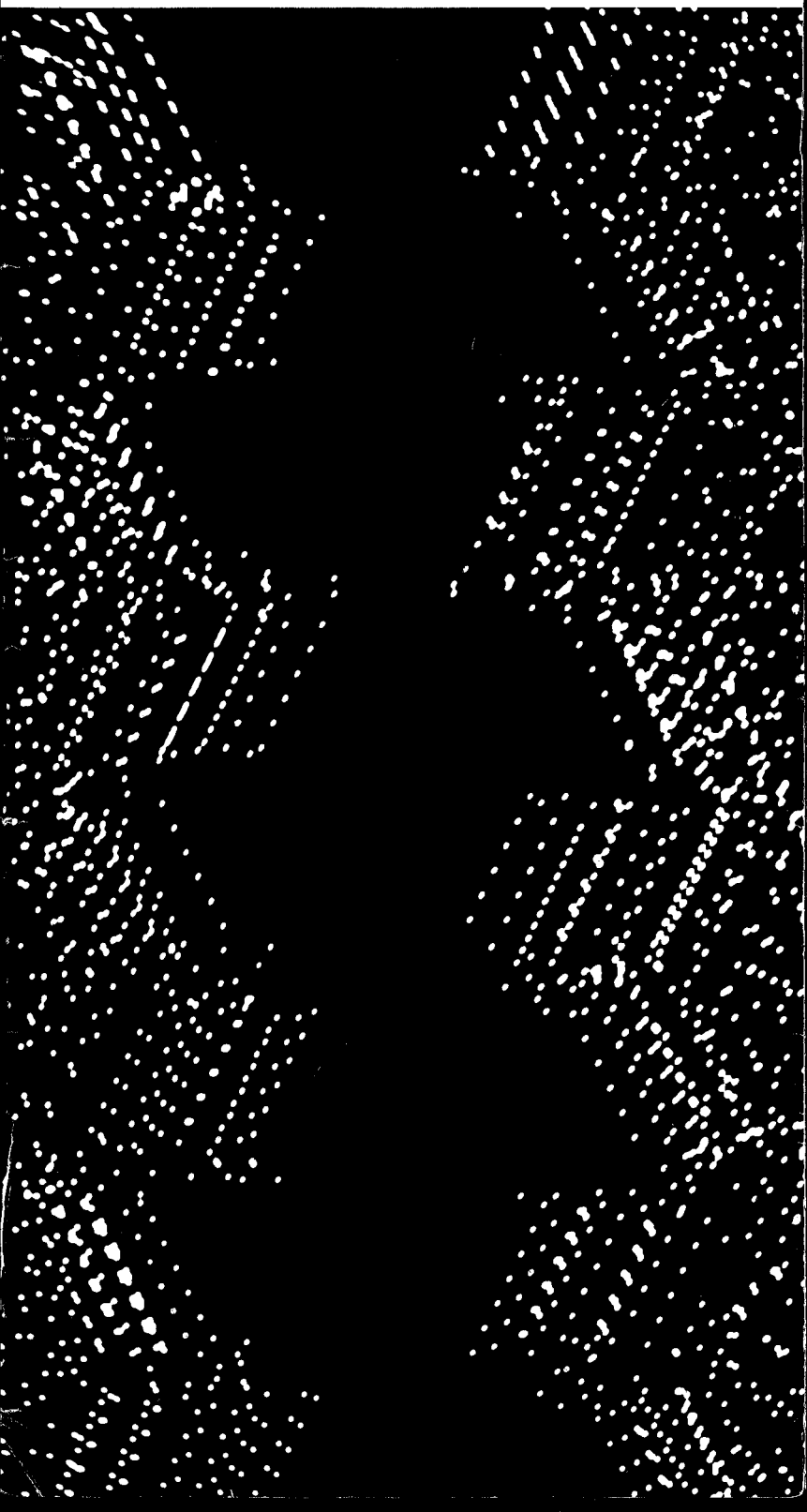
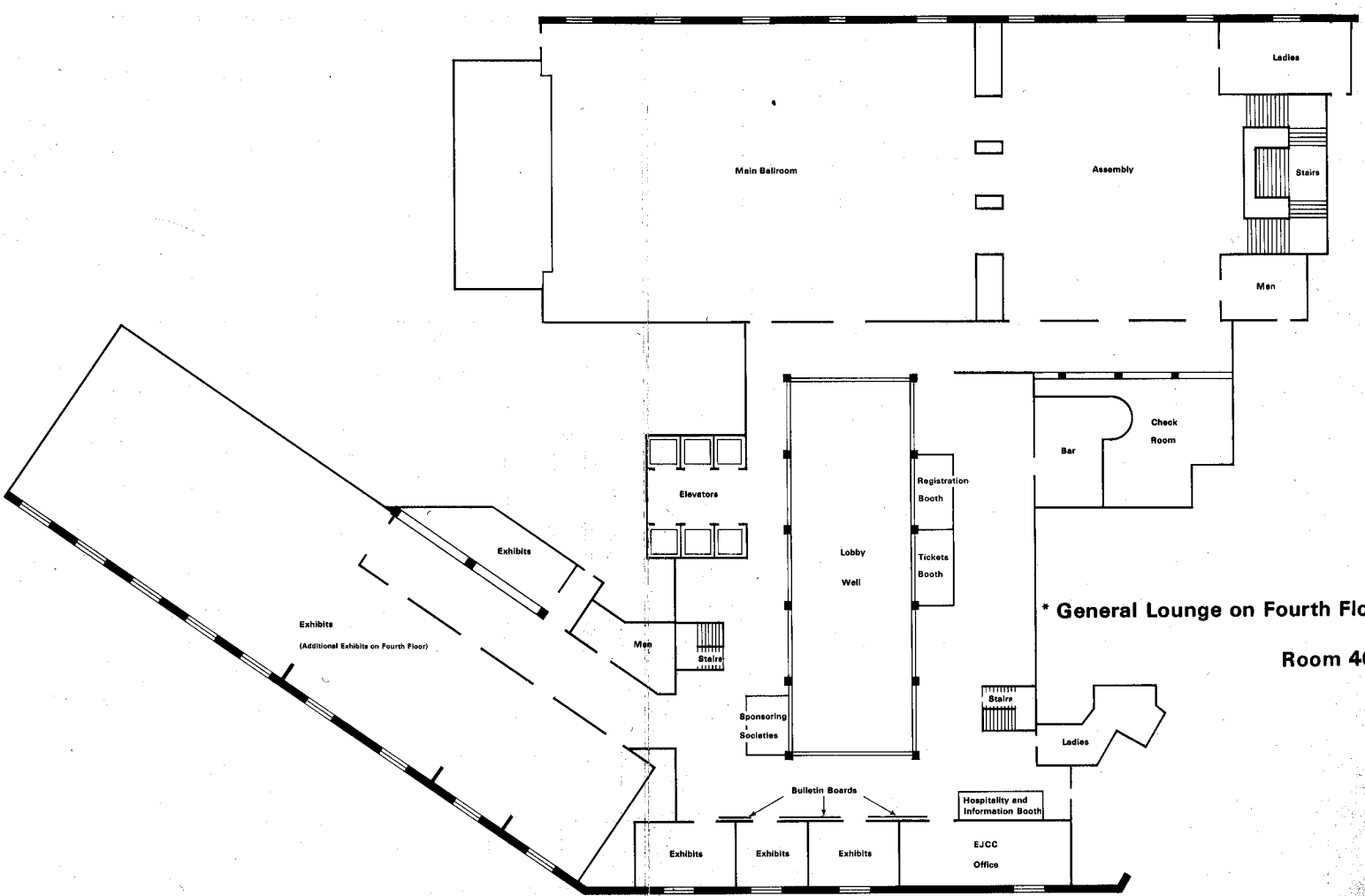
1959 Eastern Joint Computer Conference and Exhibition





* General Lounge on Fourth Floor, Room 400

Mezzanine

1959 Eastern Joint Computer Conference and Exhibition

December 1, 2, and 3, 1959
Statler Hilton Hotel
Boston, Massachusetts

Sponsored jointly by the Association for Computing Machinery, the American Institute of Electrical Engineers, and the Institute of Radio Engineers.

The entire program will take place at the Statler Hilton Hotel.

Registration will take place on the Hotel Mezzanine on Monday, November 30, 6 to 9 p.m. and Tuesday through Thursday, December 1–3, 9 a.m. to 5 p.m.

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The designs in this program were generated by the repeated use of one micro-instruction which performs arithmetic, logical, and display operations.

The computer used was the TX-0, an experimental computer developed by Lincoln Laboratory, now on loan to the Massachusetts Institute of Technology, Department of Electrical Engineering, for research and education.

1	Program
6	Abstracts (listed by session)
21	Group Discussions
25	Cocktail Party and Dinner
26	Exhibits
27	Index of Exhibitors
29	Trips
30	Hospitality
31	Miscellaneous Information

Program

Tuesday, December 1, 1959

10:00 am–12:00 n

Main Ballroom

Session 1

Conference Welcome
F. E. Heart, Chairman *1959 EJCC*
Lincoln Laboratory



Computers of the Future
R. Rice, *IBM*

Negative-Resistance Elements As Digital Computer Components
M. H. Lewin, *RCA Laboratories*

2:00 pm–5:00 pm

Main Ballroom

Session 2

H. Aiken, *Session Chairman,*
Harvard Computation Laboratory

Deposited Magnetic Films as Logic Elements
A. Franck, G. Marette, and B. Parsegyan,
Remington Rand Univac

Solid State Microwave High Speed Computers
J. A. Rajchman, *RCA Laboratories*

The Engineering Design of the STRETCH Computer
E. Bloch, *IBM*

Design of the LARC System
Part I
J. P. Eckert, J. C. Chu, and A. B. Tonik,
Remington Rand Univac
Part II
H. Lukoff, L. M. Spandorfer, and
F. F. Lee, *Remington Rand Univac*

6:00 pm–8:00 pm
Main Ballroom

EJCC Cocktail Party

Wednesday, December 2, 1959

9:00 am–12:00 n

Main Ballroom

Session 3

M. Rubinoff, *Session Chairman, Moore School of Electrical Engineering*

Control and Arithmetic Techniques in a Multi-Programmed Computer

N. Lourie, H. Schrimpf, R. Reach, and W. Kahn, *Datamatic Division, Minneapolis-Honeywell Regulator Company*

The Virtual Memory in the STRETCH Computer

J. Cocke and H. G. Kolsky, *IBM*

A Combined Analog-Digital Differential Analyzer

H. K. Skramstad
National Bureau of Standards

The System Organization of MOBIDIC B

S. K. Chao, *Sylvania Electric Products*

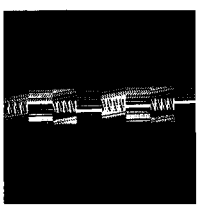
A Universal Computer Capable of Executing an Arbitrary Number of Sub-Programs Simultaneously

J. Holland, *University of Michigan*

2:00 pm–5:00 pm

Main Ballroom

Session 4



P. Armer, *Session Chairman, The RAND Corporation*

The Multi-Sequence Computer as a Communications Tool

J. N. Ackley,
International Electric Corporation

Synthesis of Switching Two Terminals Based on the Theory of G. R. Kirchhoff and O. Veblen

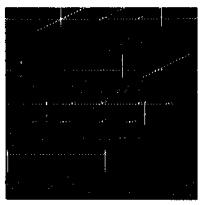
S. Okada, *Brooklyn Polytechnic Institute*

Applications of Boolean Matrices to the Analysis of Flow Diagrams

R. T. Prosser, *Lincoln Laboratory*

SIMCOM—The Simulator Compiler

T. Sanborn, *Space Technology Laboratories*



Techniques and Methods Employed in a Digital Computer Program to Solve General Transient Heat Transfer Problems

D. J. Campbell and Mrs. D. Vollenweider, *General Electric*

8:00 pm

Group Discussions

Room 400
Fourth Floor

Large Signal Equivalents in the Analysis of Circuit Tolerance

Panelists

E. U. Cohler, *Sylvania Electric Products*
R. J. Domenico, *IBM*
J. G. Linvill, *Stanford University*
T. A. Murrell, *University of Illinois*
R. L. Pritchard, *Texas Instrument*
W. J. Poppelbaum, *University of Illinois*

Room 401
Fourth Floor

The Role of Computers in the Development of Digital Systems

Panelists

T. H. Crowley, *Bell Telephone Laboratories*
P. W. Case, *IBM*
W. Gordon, *Datamatic Division, Minneapolis-Honeywell Regulator Company*
J. P. Malbrain, *Ramo-Wooldridge*
A. L. Leiner, *National Bureau of Standards*

Room 402
Fourth Floor

System Aspects of the Utilization of Kilomegacycle Components

Panelists

R. A. Kudlich, *AC Spark Plug*
M. C. Andrews, *IBM*
J. P. Eckert, *Sperry Rand*
S. Frankel, *Consultant*
D. L. Hogan, *Department of Defense*
R. E. Meagher, *University of Illinois*

Room 436
Fourth Floor

Judicious Use of Your Computer

Panelists

J. A. Buckland, *Shell Oil Company*
D. L. Johnson, *University of Washington*
F. J. Maginniss, *General Electric*
L. N. McClung, *Johns Hopkins University*
H. W. Richmond, *System Development Corporation*
H. Semarne, *Douglas Aircraft Company*

Program

Thursday, December 3, 1959

9:00 am–10:35 am **Session 5**

Main Ballroom

A. L. Samuel, *Session Chairman, IBM*

The Automatic Transcription of Machine Shorthand

G. Salton, *Harvard University and Sylvania Electric Products*

Critical-Path Planning and Scheduling

J. E. Kelley, Jr. and M. R. Walker, *Mauchly Associates*

The Automatic Digital Computer as an Aid in Medical Diagnosis

C. B. Crumb, Jr., *Bendix Systems Division;* and C. E. Rupe, M.D., *Henry Ford Hospital*

10:55 am–12:30 pm **Session 6**

Main Ballroom

W. N. Papian, *Session Chairman, Lincoln Laboratory*



An Advanced Magnetic Tape System for Data Processing

R. B. Lawrance, *Datamatic Division, Minneapolis-Honeywell Regulator Company*

A High Speed, Small Size Magnetic Drum Memory Unit for Ultra Small Digital Computers

R. A. Howard, M. May, G. Miller, and G. Shifrin, *Ramo-Wooldridge*

Temperature Compensation for a Core Memory

A. Ashley, E. Cohler, and W. S. Humphrey, *Sylvania Electric Products*

2:30 pm–5:00 pm **Session 7**

Main Ballroom

O. G. Selfridge, *Session Chairman, Lincoln Laboratory*



Use of a Computer to Design Character Recognition Logic

R. J. Evey, *IBM*

A Self Organizing Logical System

R. L. Mattson, *Lockheed Missiles and Space Division*

Alpha-Numeric Character Recognition Using Local Operations

J. S. Bomba, *Bell Telephone Laboratories*

Pattern Recognition and Reading by Machine

W. W. Bledsoe and I. Browning, *Sandia Corporation*

Discussion of Problems in Pattern Recognition

Authors
O. G. Selfridge, *Lincoln Laboratory*
U. Neisser, *Brandeis University*
R. Kirsch, *Bureau of Standards*
M. Minsky, *Lincoln Laboratory*

6:30 pm
Room to be Announced

Dinner and Award Presentation

Dr. W. H. Ware, *Guest Speaker, The RAND Corporation*

Status of Computer Developments in the Soviet Union

Award Presentation

J. H. Felker, *Program Chairman, A.T.&T.*

7 Room 439 Publ Committee 5:15-5:45 8:00-11:00?

Session 1
Tuesday

Computers of the Future

R. Rice, *IBM*

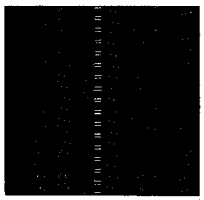
This discussion is concerned with a radical change in the technology utilized to manufacture digital data processing systems. A picture of the effect of this change on our way of specifying and designing systems is presented.

Present methods of circuit-system standardization are contrasted with anticipated future methods. An illustrative example of "System Function" design and "System Tailored" circuits and devices is given. A summary is made of the more important work required in order to progress from present to desired future systems.

Session 1
Tuesday

Negative-Resistance Elements as Digital Computer Components

M. H. Lewin, *RCA Laboratories*



The use of two-terminal negative-resistance devices as the basic switching elements in a digital system is discussed. Two fundamental problems analyzed are concerned with:

1 Achieving logical gain at maximum possible repetition rate.

2 Providing means to dictate the direction of flow of information in the system.

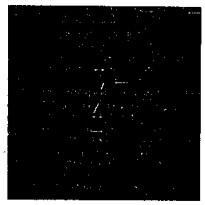
Circuits performing all the essential logical functions are presented, utilizing the "tunnel (Esaki) diode," a new high-speed negative-resistance semiconductor device, as the basic element. Single-ended and balanced circuit configurations are discussed. In addition, simple arrangements of a small number of tunnel diodes are derived to realize more complicated logical functions.

Part of the system described is a three-phase pulse power supply. Utilizing such a power source, all storage functions can be realized by dynamic storage techniques.

Session 2
Tuesday

Deposited Magnetic Films as Logic Elements

A. Franck, G. Marette, and B. Parsegyan, *Remington Rand Univac*



Deposited magnetic films have exhibited logical properties that portend their use as other than memory elements in computer applications. These logical modes of operation are explained as they pertain to a logical device, specifically a scale factoring circuit.

The principles of array logic are defined wherein multiple copies of a word are used along with the proper arrangement and selection of sense lines linking these words to obtain the desired result in a single operation rather than a series of sequential operations. The advantages of deposited magnetic films in this application are enumerated in detail in the paper.

Session 2
Tuesday

Solid State Microwave High Speed Computers

J. A. Rajchman, *RCA Laboratories*

Two types of semiconductor devices offer possibilities to speed computer rates up to a thousand megacycles.

1 Variable capacity diodes in parametric subharmonic phase locked oscillators have permitted pumping up to 10 kmc and making of 4 kmc-pumped 100 mc logic circuits. Junction type diodes of order of magnitude higher speed capability were developed.

2 Tunnel diodes switching in less than 10^{-9} seconds were developed. Scaled-down frequency logic pulse type circuits were demonstrated. Arrays of tunnel diodes promise random access memories with access cycles of 10^{-8} to 10^{-7} seconds.

Both diodes were developed in microcapsules fitting within microwave transmission boards and dissipate sufficiently low power to permit necessary high packing density for ultra fast computers. Special circuit techniques for use of two terminal single port devices were developed.

Session 2
Tuesday

The Engineering Design of the STRETCH Computer

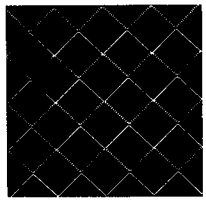
E. Bloch, *IBM*

The engineering design of the STRETCH Computer is described, emphasizing the detailed internal organization of the major units and systems, such as Indexing units, the Floating Point Arithmetic units, the Lookahead, and the Memory bus. The logical circuit and general package is briefly mentioned.

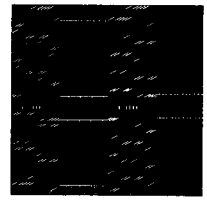
Session 2
Tuesday

Design of the LARC System

Part I J. P. Eckert, J. C. Chu, and A. B. Tonik, *Remington Rand Univac*



The LARC specifications are reviewed to point out how closely the system constructed conforms to the specifications as announced several years ago. The various levels of memory that are available in the system are described. The advantages of the particular arrangement of the memories employed are contrasted to other possible memory arrangements which could have been employed but which would have been difficult to program and would have limited the flexibility of the system. The speeds of operation are discussed along with how these speeds are achieved by overlapped operation and parallelism. The controlling of several semi-independent programs as well as input-output operations are discussed.



Part II H. Lukoff, L. M. Spandorfer, and F. F. Lee, *Remington Rand Univac*

The method of selection of the circuitry is discussed and is based on an optimization of cost, speed and logical capabilities. The fabrication of these circuits into an operating system is then covered with emphasis on the connector design, packaging considerations, noise problems and maintenance features. Success of the system was dependent upon development of packaging techniques that permitted very high packing densities. Mechanized backboard wiring techniques were used to achieve highest circuit speeds. Fine steel core wire is used for all wiring, which required the development of handling techniques. Methods of working with high component densities are discussed. Solid state regulated power supplies provide reliable operation.

The present status of the system is discussed.

Session 3
Wednesday

Control and Arithmetic Techniques in a Multi-Programmed Computer

N. Lourie, H. Schrimpf, R. Reach, and W. Kahn, *Datamatic Division, Minneapolis-Honeywell Regulator Company*

This paper describes how a 256-word 20-bit coincident current memory, operating out of phase with the main storage memory, can be efficiently utilized to aid in the performance of relatively complicated control functions in a digital computer with no loss in machine speed. These functions include such things as 16 sequencing counters to automatically control up to 8 independent programs, buffering controls for 64 peripheral devices, and indexing and tabular control of main memory addresses.

computing / sec / prob
6 μs mem cycle time

A parallel-serial parallel bit configuration of data which allows some time sharing of the adder circuitry is described. The resultant speed approaches that of an all parallel system, but requires not nearly as much equipment. It is easily adapted to perform addition and subtraction in either binary or decimal codes.

Session 3
Wednesday
(2)

The Virtual Memory in the STRETCH Computer

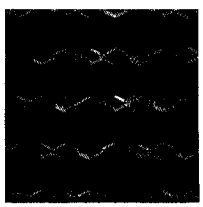
J. Cocke and H. G. Kolsky, *IBM*

The use of high speed logical devices and more sophisticated design methods has resulted in an imbalance between the arithmetic and memory speeds. The Virtual Memory provides a method of "looking-ahead" in order to permit the simultaneous operation of several blocks of memory in an asynchronous computer while maintaining logical correctness. Machine speed is further increased by overlapping the bookkeeping operations with the principal computations. To measure the effectiveness of the Virtual Memory, a timing simulation program for STRETCH was written on the IBM 704. This program enables us to study a wide variety of parameters for design optimization.

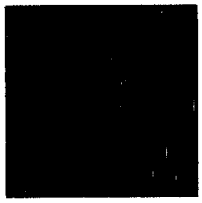
Session 3
Wednesday
(1)

A Combined Analog-Digital Differential Analyzer

H. K. Skramstad, *National Bureau of Standards*



An analog-digital differential analyzer has been designed which combines the analog advantages of high speed and continuous representation of variables with the digital capability of high precision and dynamic range. It is based on representing dependent variables by two quantities, a digital number representing the more significant part, and an electrical voltage representing the less significant part. As in the electronic analog computer, time is the independent variable.



The design of components required to build a computer of this combined type, such as integrators, summers, and multipliers are given, and examples of how the solution of a few elementary differential equations would be carried out are presented.

Session 3
Wednesday
(4)

The System Organization of MOBIDIC B

S. K. Chao, *Sylvania Electric Products*

2 central processors
38 bits
50 M bit disk
8 disks
(34 1/2" dia)
4096 tracks
Serial Serial
continuous loop
8 μs Mem
1 Mc clock

MOBIDIC B is an all transistorized, militarized computer mounted in a standard Army trailer. It is a general purpose, parallel, binary, synchronous, fixed point, and duplexed data processing system.

It contains two basic processors, identical in characteristics, internally tied together to the same system transfer bus. Both processors share a common set of input-output devices. Each processor is capable of operating an independent program without interference. They are also capable of duplexed operation, allowing either processor to monitor and exert control over the other.

In addition to the 8192-word high speed Core Memory in each processor, there exists a 50-million-bit Mass Memory. This memory is treated as an input-output device, addressable by in-out instructions. A Data Retrieval Unit is incorporated to facilitate data searching from the Magnetic Tape and Mass Memory.

Session 3
Wednesday

A Universal Computer Capable of Executing an Arbitrary Number of Sub-Programs Simultaneously

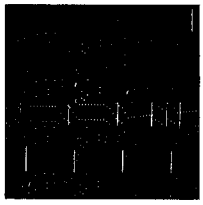
J. Holland, *University of Michigan*

The paper describes a universal computer capable of simultaneously executing an arbitrary number of sub-programs, the number of such sub-programs varying as a function of time under program control or as directed by input to the computer. Three features of the computer are:

1 The structure of the computer is a two-dimensional modular (or iterative) network.

2 Each sub-program is spatially organized, thus facilitating the simulation of "highly-parallel" systems having many points or parts.

3 The computer's structure and behavior can, with simple generalizations, be formulated so as to make it a useful abstract tool for investigating problems in automata theory.

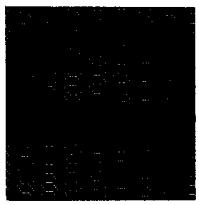


Session 4
Wednesday

The Multi-Sequence Computer as a Communications Tool

J. N. Ackley, *International Electric Corporation*

This paper describes possible applications, as a communications tool, of a multi-sequence computer in which more than one sequence or program operates independently, time-sharing the central processing unit. The computer is made to time-share on an on-demand basis between all of the input and output devices. The control sequences and the buffering can be provided by the central



processing unit. A multi-sequence computer, which permits integration of a multiplicity of input and output devices economically, becomes a very rapid and economical message switching center by connecting the communications lines as the input and output devices. This configuration can also be exploited as a real time data processing system or a real time control system.

Session 4
Wednesday

Synthesis of Switching 2-Terminals Based on the Theory of G. R. Kirchhoff and O. Veblen

S. Okada, *Brooklyn Polytechnic Institute*

A routine procedure for determining minimum switching networks, non-series-parallel and non-single-contact in general, for any given Boolean polynomial R is exemplified.

1 Determine a generating system of the invariant transformation group of R.

2 Express R in a cut-set matrix B^{t}_{k} or in a tie-set matrix C_{g}^{k} and reduce it to linearly independent sets B^{a}_{k} or C_{q}^{k}.

3 Conversely, determine all cut-sets of B^{a}_{k} or tie-sets of C_{q}^{k} by linear combination.

4 If B^{a}_{k} or C_{q}^{k} has an over-set, increase contacts and repeat (2) and (3) until over-sets disappear.

5 Realize B^{a}_{k} or C_{q}^{k}. If it is nonrealizable, increase contacts and repeat (2)-(5).

Session 4
Wednesday

Applications of Boolean Matrices to The Analysis of Flow Diagrams

R. T. Prosser, *Lincoln Laboratory*

An analysis of the structure of flow diagrams, such as those associated with computing machine programs, can be given in terms of Boolean matrices. With each such diagram is associated a pair of Boolean matrices. The first of these, called the connectivity matrix, contains the topological structure of the

diagram, and the second, called the precedence matrix, contains its precedence relations. Elementary computations on these matrices are shown to yield detailed information concerning the internal logical consistency of the flow diagram. Possible applications to automatic debugging procedures are suggested.

Session 4
Wednesday

SIMCOM—The Simulator Compiler

T. Sanborn, *Space Technology Laboratories*



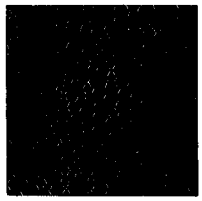
SIMCOM is a specialized compiler used in preparing computer-simulation programs. The input language consists of stylized "sentences" which specify the configuration of the computer to be simulated, and describe the bit-wise effect of each of the simulated computer's operations. Related sentences are grouped into "paragraphs" to minimize coded cross references in the simulation program. Because the compiler output is a symbolic code, it can be generated during a single pass through the source program. This symbolic code is then processed by a conventional two-pass assembly program. Automatic storage allocation, and a novel form of subroutine add to the power of the system.

Session 4
Wednesday

Techniques and Methods Employed in a Digital Computer Program to Solve General Transient Heat Transfer Problems

D. J. Campbell and Mrs. D. Vollenweider, *General Electric*

A method is presented for solving transient and steady state heat transfer problems. The digital computer program permits the analysis of three-dimensional problems with arbitrary geometry and



various combined modes of heat transfer. An unusual manner of describing geometric configurations and of presenting input data in symbolic form is discussed. Aspects of input processing, testing, and organization are described. Further extensions of these methods and computational techniques are proposed.

Session 5
Thursday

The Automatic Transcription of Machine Shorthand

G. Salton, *Sylvania Electric Products*

The transformation of speech input into written output is an important part of many data processing problems. Since digitalized speech input cannot be readily recognized, it is suggested that machine shorthand notes be used as input to a computer process for the automatic production of written output.

The basic theory of machine shorthand is briefly described. Thereafter, various methods for the transcription of machine shorthand are discussed. Many problems arising in the translation process are found to be similar to problems of interest in the machine translation of languages and in the analysis of speech. Computer routines for the production of comprehensible, although not syntactically or semantically unambiguous, English are described, and computer output is demonstrated.

Session 5
Thursday

Critical-Path Planning and Scheduling

J. E. Kelley, Jr. and M. R. Walker, *Mauchly Associates*

Since January 1957 a technique for planning, scheduling, and coordinating complex engineering-type projects has been developed, tested, and applied. The essential ingredient of the technique is a working model of a project that incorporates sequence information, costs, and durations for each component of the project. Parametric linear

programming is used to combine these factors to produce optimal cost schedules having varying completion dates. Included is a complete system for cost accounting and computing and updating all project information. The rudiments of the technique, the history of the study, and application results are discussed.

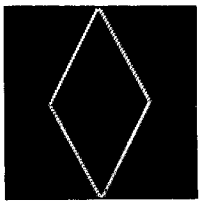
Session 5
Thursday

The Automatic Digital Computer as an Aid in Medical Diagnosis

C. B. Crumb, Jr., *Bendix Systems Division;* and C. E. Rupe, M.D., *Henry Ford Hospital*

The idea presented in this paper is that an automatic digital computer can provide great assistance to the medical diagnostician by rapid calculation, based upon symptoms and physical findings, of the relative probability that a certain physical disorder fits that set of findings. The plan calls for use of the computer to prepare statistical tables of correlation constants between patients' symptoms and various associated medical disorders; and to compute relative probabilities that various disorders fit a given set of symptoms.

The paper describes the origins of the scheme, the plan for development of the technique, and progress to date in execution of that plan.

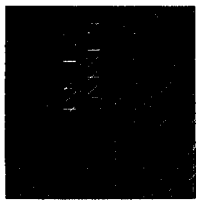


Session 6
Thursday

An Advanced Magnetic Tape System for Data Processing

R. B. Lawrance, *Datamatic Division, Minneapolis-Honeywell Regulator Company*

This paper describes a new magnetic tape mechanism, recording system, and information checking and restoration system of high reliability. In the mechanism, comparison is made between pinch-roller and vacuum capstan means



for tape motion control. Criteria include minimization of tape deterioration, both gradual and catastrophic; tracking skew and maintenance considerations.

Systems techniques for enhancing tape system reliability are discussed briefly, with some emphasis on the use of error detection and automatic correction. The choice of information format on the tape and of error-correcting parameters for maximum effectiveness is described.

Session 6
Thursday

A High Speed, Small Size Magnetic Drum Memory Unit for Ultra Small Digital Computers

R. A. Howard, M. May, G. Miller, and G. Shifrin, *Ramo-Wooldridge*

A magnetic memory drum has been designed and constructed for use in future aircraft and missile weapons systems. The drum has the following significant characteristics:
Over-all Size—4" x 4" x 6"
Memory Capacity—300,000 bits stored on 122 tracks.
Operating Bit Rate—546 kc.
Speed—12,000 rpm.
Ten circulating registers with a minimum length of 26 bits are also provided.

The drum is ruggedly built to withstand MIL-E-5400 Class 2 specifications. It can stand shock loads of 16g and continuous vibrations of 5g peak from 0–2000 cps without the use of shock mounts.

Session 6
Thursday

Temperature Compensation for a Core Memory

A. Ashley, E. Cohler, and W. S. Humphrey, *Sylvania Electric Products*

For fixed installation, it is often possible to control the temperature of ferrite core memories within narrow limits. However, in a small mobile computer de-

signed to operate over world-wide conditions, this control is not feasible because of the added weight, volume, and cost encountered. A memory designed for such application has been temperature compensated by the use of temperature sensitive components in the current sources to the x-y drivers and in the power supplies for the z drivers. In addition, core-derived strobing has provided peaking time compensation for the sense amplifiers as changes in transistor characteristics delay or advance drive current. This compensation permits operation of an 8192-word 38-bit transistorized memory running at an 8-microsecond cycle time in an ambient environment which may vary between —30° C and +55° C.

Session 7
Thursday

Use of a Computer to Design Character Recognition Logic

R. J. Evey, *IBM*



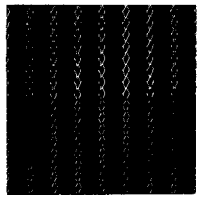
Character recognition logic for the IBM 1210 Sorter/Reader was developed with the aid of an IBM 704 computer. Characters printed with magnetic ink are quantized by the Sorter's scanning system into a 7 x 10 binary matrix which drives recognition logic of the AND and OR type. To develop this logic, computer programs which simulated the quantizing and the logic were used in conjunction with hardware that provided "real-life" character degradation. These programs and the procedure for developing the recognition logic are discussed.

Session 7
Thursday

A Self Organizing Logical System

R. L. Mattson, *Lockheed Missiles and Space Division*

A self-organizing logical system can be thought of as two systems: one, a network of adjustable logical devices which receives binary input combinations and



operates on these to produce a binary output; the other, a system for determining what adjustments should be made in the logical devices to produce a desired logical function. By using the performance of the logical devices as a feedback to the adjusting system, optimum logical function synthesis can be obtained.

In this paper a precise model for an adjustable logical device and a convergent iterative adjustment procedure is given and analyzed. Four demonstrations are given which make use of these logical devices and iterative procedure to synthesize an optimum function for pattern recognition problems containing from 25 to 216 binary variables. These problems were simulated on the IBM 704 computer, and each problem required less than two minutes of computer time to synthesize the optimum truth function for the process considered.

Session 7
Thursday

Alpha-Numeric Character Recognition Using Local Operations

J. S. Bomba, *Bell Telephone Laboratories*

A demonstration of the recognition of 34 distinguishable, handprinted, block capital, alpha-numeric characters is described. The basis of the recognition method is the use of "local operations" to extract significant features from the characters. The recognition procedure is:

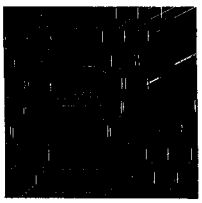
1 Reduce the noise in the field by local averaging.
2 Standardize the character line width.
3 Extract the "essential" features from the character (typical features are straight lines of varying slopes and line intersections).
4 Derive the relative locations and sizes of these features.
5 Identify the characters from items. (3) and (4) with combinational logic. The method was successfully evaluated with a simulation program on an IBM 704.

Session 7
Thursday

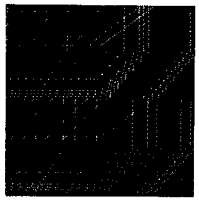
Pattern Recognition and Reading by Machine

W. W. Bledsoe and I. Browning, *Sandia Corporation*

An essential part of data processing by machine is pattern discrimination, characterization, and mensuration. To accomplish this end, a means was devised and is being used to so treat patterns with an IBM 704. As a means of evaluating the discriminating capability of the system, typewritten numerals, hand-blocked print, and hand-written script were used as patterns of respectively increasing complexity and individual variability.



All of these patterns were recognized by a single, general-purpose program from the initial patterns which were optically imaged on a 10 x 15 photocell matrix. Recognition was accomplished by individual character and by contextual relationships.



Four small group discussions will be held on Wednesday evening, December 2 at 8:00 p.m. Each discussion will be led by a group of distinguished panelists, expert in their fields. Members of the audience will be able to exchange opinions with one another and with the discussion leaders. Pertinent information for each discussion group is given below.

Room 400
Fourth Floor

Large Signal Equivalents in the Analysis of Circuit Tolerance

Panelists:

E. U. Cohler, *Sylvania Electric Products*
R. J. Domenico, *IBM*
J. G. Linvill, *Stanford University*
T. A. Murrell, *University of Illinois*
R. L. Pritchard, *Texas Instrument*
W. J. Poppelbaum, *University of Illinois*

The object of the panel discussion is to have some experts in the field air their views concerning:

1 The physicists' approach to equivalent circuits for switching applications.
2 The models actually used by organizations designing computers.

Since the first group tends to think in terms of quite complex models, while the second group has to use the simplest possible equivalent representation, it is anticipated that considerable discussion will take place. Hopefully the result will be the acceptance of practically useful equivalents for transistors and diodes both for dc and transient analysis.

Room 401
Fourth Floor

The Role of Computers in the Development of Digital Systems

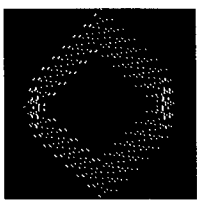
Panelists:

T. H. Crowley, *Bell Telephone Laboratories*
P. W. Case, *IBM*
W. Gordon, *Datamatic Division, Minneapolis-Honeywell Regulator Company*
J. P. Malbrain, *Ramo-Wooldridge*
A. L. Leiner, *National Bureau of Standards*

The use of general purpose digital computers to aid in the development of digital systems has expanded greatly in recent years. A number of companies have announced the availability of computer programs to do a portion of the routine engineering of new systems. Many companies are presently expanding such programs into something approaching an integrated system to carry out such functions as:

1 Performance of simple design jobs.
2 Simulation and checking of manual designs.
3 Record keeping and preparation of drawings.
4 Facilitation of changes.

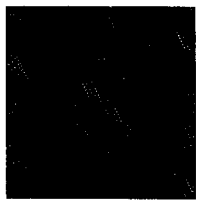
In many aspects of this problem, different companies appear to be using similar techniques to achieve the same objectives. The panel will attempt to bring out those areas which involve special objectives or novel techniques. Panel members will discuss such topics as:



1 Comparison of existing programs.
2 Summary of experience with such programs.
3 Advantages and disadvantages.
4 Probable future extensions.

Room 402
Fourth Floor

System Aspects of the Utilization of Kilomegacycle Components



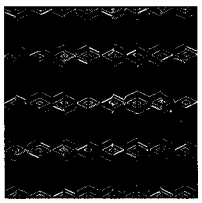
Panelists:

R. A. Kudlich, *AC Spark Plug*
M. C. Andrews, *IBM*
J. P. Eckert, *Sperry Rand*
S. Frankel, *Consultant*
D. L. Hogan, *Department of Defense*
R. E. Meagher, *University of Illinois*

Work now being done by various organizations gives promise that before very long we may have components which are able to perform in the order of a billion logical operations a second.

The panel will discuss such problems as system organization, provision of storage facilities with suitable access speeds, input and output arrangements, signal propagation delays, heat dissipation, initial debugging and maintenance techniques, and appropriate applications for such systems.

Room 436
Fourth Floor



Judicious Use of Your Computer

Panelists:

J. A. Buckland, *Shell Oil Company*
D. L. Johnson, *University of Washington*
F. J. Maginniss, *General Electric*
L. N. McClung, *Johns Hopkins University*
H. W. Richmond, *System Development Corporation*
H. Semarne, *Douglas Aircraft Company*

The general purpose computer is no longer a novelty in industry. Its use at each industrial organization determines whether it is an economic tool or an expensive toy.

How are you using your computer?

Has a computer encouraged your organization to simplify computing procedures, to redefine old problems, and to investigate new endeavors?

Are your computer applications based primarily on computer availability or on problem requirements?

Does a knowledge of programming techniques influence the analysis of the problem?

Has automatic programming extended the range of new applications?

How much original research is attempted and required for each new problem?

Do the engineers use the systems approach or the Topsy method when attacking a new computer application?

Has the solution or an intermediate result promoted further related studies?

What are some of the fringe benefits for the engineer and manager?

Leaders in the computing field have been invited to discuss these and similar questions on applications with the engineers, programmers, managers, analysts, and others.

Cocktail Party and Dinner



For your social enjoyment, the EJCC has arranged two special functions: a Cocktail Party to get acquainted, and a Dinner to conclude the Conference. Wives are cordially invited to attend both functions. Tickets for these events may be obtained at the Tickets Booth on the Mezzanine of the Statler Hilton Hotel.

Reception and Cocktail Party

Tuesday evening, December 1,
6 to 8 p.m.
Main Ballroom
Tickets $4.50 per person

Dinner and Award Presentation

Thursday evening, December 3,
6:30 p.m.
Room to be announced
Tickets $7.50 per person

Status of Computer Developments in the Soviet Union

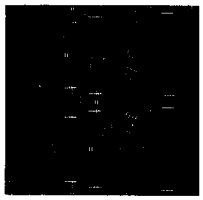
Dr. W. H. Ware, *The RAND Corporation*

In the summer of 1959 a delegation sponsored by the National Joint Computer Committee visited, in the Soviet Union, a number of research institutes, universities, and factories producing digital computers. This delegation received the most detailed picture of Soviet developments yet given to a group from the West. Dr. Ware was a member of the delegation and editor of the delegation's report.

Award Presentation

J. H. Felker, *Chairman, Program Committee, AT & T*

In recognition of the fact that technical programs are sometimes marred by careless or obtuse presentation of papers, the EJCC has decided to emphasize the importance of a good oral presentation. An award of $300 will be made for the best presentation at the Conference of a paper describing significant work in the computer field. The winner of the award will be selected by the Program Committee and the presentation will be made at this dinner.



Exhibits of the latest developments in complete data-processing and electronic computer systems, computer components, and sub-assemblies are on the Mezzanine and Fourth Floor of the Statler Hilton Hotel. The Conference Committee cordially invites you to bring your personal friends, business associates, and customers to this comprehensive display. There is no admission charge.

Exhibit Hours:

Tuesday, December 1,
10 a.m. to 5:30 p.m.

Wednesday, December 2,
9:30 a.m. to 9 p.m.

Thursday, December 3,
9:30 a.m. to 5:30 p.m.

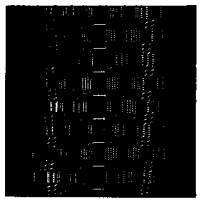
Location*	
M 27, 28	Aeronutronic, Division of Ford Motor Company
M 34, 35	AMP, Inc.
M 51, 52	Ampex Corporation
403	ANelex Corporation
M 46, 47, 48	Autonetics, Division of North American Aviation, Inc.
M 44, 45	Bendix Computer Division
411	Benson-Lehner Corporation
M 17	Bryant Computer Products Division
M 30, 31, 32	Burroughs Corporation, ElectroData Division
M 53	California Computer Products, Inc.
M 18, 19	C-E-I-R, Inc.
412	C & K Components, Inc.
M 33	C. P. Clare & Company
M 12, 13	Computer Control Company, Inc.
414	Di/An Controls, Inc.
M 41, 42	Digital Equipment Corporation
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414	Fairchild Semiconductor Corporation
403	Ferranti Electric, Inc.
413	General Ceramics Corporation
M 14	General Electric Company, Light Military Electronics Dept.
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M 54	Harford Metal Products, Inc.
410	Harvey-Wells Electronics, Inc.
M 55	Instrument Specialties Company, Inc.
M 57	Intellectronics Laboratories of the Ramo Wooldridge Division of Thompson Ramo Wooldridge, Inc.
M 22, 23, 24	International Business Machines Corporation

* M—Mezzanine, 400's—Fourth Floor. Maps are printed on the inside cover of this program.

M 8, 9	Laboratory for Electronics, Inc.
M 15, 16	Librascope, Inc.
M 25	Lumatron Electronics, Inc.
406	Micro Switch, Division of Minneapolis-Honeywell Regulator Company
M 39, 40	Minneapolis-Honeywell Regulator Company, DATAmatic Division
M 43	Minnesota Mining and Manufacturing Company
418	The National Cash Register Company
409	Packard Bell Computer Corporation
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415	Radio Corporation of America
403	Reeves Soundcraft Corporation
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M 29	Rese Engineering, Inc.
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414	Tally Register Corporation
407	Telemeter Magnetics, Inc.
404	Teletype Corporation
409	F. D. Thompson Publications, Inc.
M 1, 2	Union Switch & Signal Division, Westinghouse Air Brake Company
M 26	Wang Laboratories, Inc.
410	John Wiley & Sons, Inc.

* M—Mezzanine, 400's—Fourth Floor. Maps are printed on the inside cover of this program.

Trips



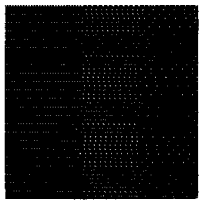
Two special trips to laboratories where advanced computer work is being carried on have been arranged for Conference registrants. Registration for these trips will be on Tuesday, December 1, at the Tickets Booth. A transportation charge of $1.50 per person will be made for each trip. Only a limited number can be accommodated.

The two special trips are:

M.I.T. Lincoln Laboratory including a tour of the TX-2 Computer on Wednesday, December 2.
This trip will be divided into two identical sections.
Section 1, 8 to 12:30 a.m.
Section 2, 1 to 5:30 p.m.
Registration for the Lincoln Laboratory trips will close at noon on Tuesday, December 1.
Because of security regulations, foreign nationals from Iron Curtain countries will be excluded from this trip.

Arthur D. Little, Inc. including a demonstration of thin-film cryotrons on Wednesday, December 2, from 6:30 to 10:15 p.m.

In addition to these trips, several organizations in the area will welcome visitors from the Conference. Details will be posted on the Bulletin Board.

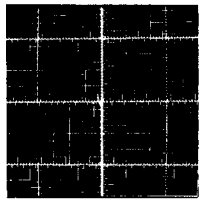


The 1959 Eastern Joint Computer Conference Committee welcomes you to historic and cultural Boston. The Committee has provided a Hospitality and Information Booth to help you find your way in Boston and make your stay more enjoyable. At this Booth you will find attendants prepared to answer your questions about the EJCC and Boston, including where to find good restaurants, entertainment, public transportation, and places of interest. Much of this information has been condensed in a pamphlet entitled "Welcome to Boston." This and other literature about Boston (including maps) are available at the Hospitality and Information Booth.

Morning Coffees for the Ladies

The Veranda—Statler Hilton Hotel
Tuesday, Wednesday, Thursday,
December 1, 2, 3, 9:30 am–11:00 am.

The EJCC has arranged morning coffees with special hostesses to help you plan each of your days in Boston. You are urged to attend all three coffees and enter in each day's activities—including sight-seeing and shopping tours, a fashion show, and other entertainment. A movie about Boston will be shown during the coffee hour on Tuesday.



The Bulletin Board on the Mezzanine will carry up-to-date information on events of the Conference.

The general registration fee includes one copy of the Conference Proceedings. Additional copies may be ordered at $3.00 per copy at the Tickets Booth or through the sponsoring societies.

Student registration does not include a copy of the Proceedings. Copies may be ordered as above.

Space for recruiting notices will be provided at the Conference Bulletin Board. A single 4" x 6" space will be available for each organization or division of a company. Notices should be given to the EJCC official at the Hospitality and Information Booth for stamping and posting.

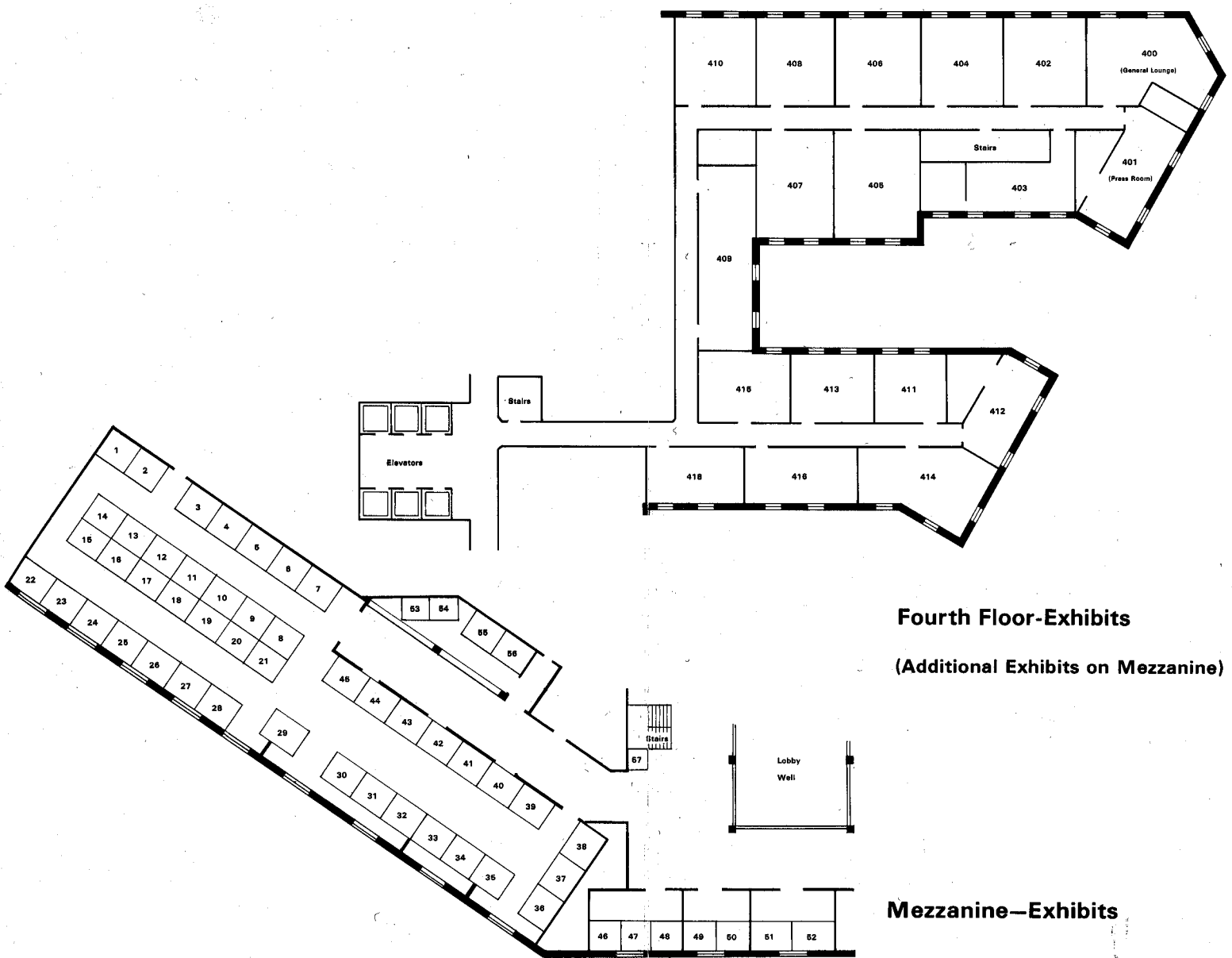
Committee Members

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Program Committee	Jean H. Felker, *Chairman; A.T.&T.* Robert A. Kudlich, *Vice Chairman, AC Spark Plug* Mandalay Grems, *IBM* Ben M. Gurley, *Digital Equipment Corporation* John W. Haanstra, *IBM* Marvin Jacoby, *Sperry Rand Corporation* W. J. Poppelbaum, *University of Illinois*
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409	Engineered Electronics Company
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Fourth Floor-Exhibits

(Additional Exhibits on Mezzanine)

Mezzanine—Exhibits

(Additional Exhibits on Fourth Floor)